A02 Technical Report: Hugging Face Transformers vs. OpenAI GPT-4 API

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WHAT WE WERE TRYING TO FIND OUT

For this project, our team wanted to answer a practical AI engineering question: if we were adding an AI feature to a real application, when would it make more sense to run a Hugging Face model and when would it make more sense to call OpenAI through an API? We compared the two approaches using usability, performance, support, scalability, control/privacy, and cost so the comparison stayed focused on decisions a development team would actually make.

WHAT OUR TEAM ACTUALLY DID

We split the work across research, the written report, the presentation, the repository, and a small sentiment-analysis demonstration. Brandon organized the presentation and demo; Elijah revised the report, managed the repository and submission structure, and led the team organization; Justin gathered and checked sources; Jonah edited the slides and checked that the presentation matched the report; and Osemudia joined during the roster confusion, coordinated with the team, reviewed the current project materials and submission structure, and helped confirm team participation before submission. For the demo, the Hugging Face side used a pretrained sentiment-analysis pipeline, while the OpenAI side used an explicit sentiment-classification prompt sent to an OpenAI model through the API. We used the demo to show the difference in workflow and setup; we did not claim numerical accuracy or speed benchmarks that we did not measure.

WHAT WE FOUND

Area	Hugging Face	OpenAI API
Usability	More setup; model and environment are chosen by the developer	Quicker start; application sends requests to a hosted model
Performance	Depends on the selected model and available hardware	Model is hosted by OpenAI; application depends on network/API access
Support	Strong open-source community and documentation	Official platform documentation and vendor-managed service
Scalability	Serving and infrastructure are the team's responsibility	Model hosting is managed by the API provider
Control / Privacy	More control over model files and where they are deployed	Less model-level control; data is sent to an external service
Cost	Hardware, hosting, setup, and maintenance can matter	Usage-based API costs instead of operating the model yourself

REAL-WORLD EVIDENCE WE USED

JPMorganChase: its AI Research program lists foundation models for the financial domain and multimodal document processing and analysis as research focus areas. That gave us a real example of why organizations working with large amounts of financial information invest in specialized AI research. **Duolingo:** OpenAI's official Duolingo case study explains that GPT-4 was used in Duolingo Max for Role Play and Explain My Answer, which gave us a clear example of a company using a managed model to add conversational AI and feedback without building the foundation model itself.

WHAT WE LEARNED

The biggest thing we learned is that Hugging Face and OpenAI are not just two tools that do the same job. They represent two different ways of building with AI. Hugging Face gives the development team more responsibility and more control over the model workflow. OpenAI removes much of that infrastructure work and lets the team focus on the application and prompting. After doing the comparison and demo, our group would not pick one automatically; we would choose based on the project's data requirements, available hardware, team experience, budget, and how quickly the application needs to be built.

SOURCES

Hugging Face, "Transformers Documentation" - https://huggingface.co/docs/transformers/index

OpenAI, "Developer Quickstart" - https://platform.openai.com/docs/quickstart

JPMorganChase, "Artificial Intelligence Research" - https://www.jpmorganchase.com/about/technology/research/ai

OpenAI, "Duolingo: Filling crucial language learning gaps" - https://openai.com/index/duolingo/